

Researchers create world's largest quantum circuit board

Researchers from the University of Tokyo and the Australian National University have reportedly developed the world's largest quantum circuit board. According to reports, this is a big step towards the development of quantum computers, which would be capable of delivering ultra-fast performance for tasks that require long and complex computation. The researchers have reportedly managed to bring 10,000 quantum systems into a single component, a huge jump from the earlier accomplished 14.

Further, reports online say that the design for the board was proposed by Nicolas Menicucci from the School of Physics at the University of Sydney. The new board is an improvement of three orders of magnitude over the earlier designs that come closer to it, which is what makes it the largest quantum circuit board in the world. Further, the researchers at the University of Tokyo were led by Professor Akira Furusawa.

Reports state that a functioning quantum computer would be able to perform the most difficult tasks and computations. It would also be able to deal with complex problems from quantum physics, which even supercomputers cannot handle at the moment.

Now a drone that can hack into other drones

Security researcher, Samy Kamkar, has reportedly developed a hacker drone, which can hijack other drones. According to reports, the Kamkar used the Parrot AR.Drone 2.0 radio-controller quadcopter embedded with a Raspberry Pi computer and two USB wireless adaptors (Alfa AWUS036H Wi-Fi network card and

Edimax EW-7811Un nani Wi-Fi adaptor).

Further, according to reports, Kamkar has developed a Perl application, which is called Skyjack running on Raspberry Pi Linux. This application can reportedly make the use of other open source software to hijack drones. The Alfa's adaptor is used by the system in order to identify other Parrot AR.Drones through their MAC addresses.

The system does this through the Aircrack-ng open source program that can hack into wireless networks. A de-authentication attack is launched on the other drone. Aircrack-ng's own documentation says it "sends disassociate packets to one or more clients which are currently associated with a particular access point."

After disconnecting the real owner from the drone, Skyjack then connects to it, whilst posing as the real owner. It then starts sending commands to the drone using another open source library called node-ar-drone. The Skyjack application can be run from almost any Linux-based laptop.

NASA's latest space technology small satellite phones home

PhoneSat 2.4, NASA's next-generation smartphone cubesat has phoned home. The tiny spacecraft that uses an off-the-shelf smartphone for a brain has completed checkout and sent back data confirming all systems are 'go' for the spy spacefarer.

PhoneSat 2.4, a cube approximately 10 centimetres square, weighs only about one kilogram, and was developed at NASA's Ames Research Center in Moffett Field, Calif. It is first of the PhoneSat family to use a two-way S-band radio, allowing engineers



PhoneSat—NASA's next-generation smartphone-enabled spacecraft

to command the satellite from Earth. It is confirming the viability of using smartphones and other commercially available electronics in satellites destined for low-Earth orbit.

"It's great to hear from NASA's most recent cubesat spacecraft," said Michael Gazarik, NASA's associate administrator for space technology in Washington. "NASA is committed to opening up the high frontier to a new generation of explorers who can take advantage of these sorts of small satellites to do science and technology development at a fraction of the cost of larger, more complex spacecraft."

In April, NASA successfully demonstrated a one-week mission with PhoneSat 1.0. With an expected orbital lifetime of up to one year, PhoneSat 2.4 will measure how well commercially developed components perform in space over a long period of time. This innovative application of commercially developed technologies for use